Job Amenities and Compensating Differentials

Labor I — Week 7

Teaching deck draft

Central question and motivation

- Week 7 asks how workers trade wages against nonwage job attributes.
- Jobs differ in safety, flexibility, predictability, commute burden, autonomy, and remote-work options.
- Wage data alone can therefore mismeasure welfare, sorting, and inequality.

Week 6 to Week 7 bridge

- Week 6 showed that family structure changes the value of time and flexibility.
- Week 7 moves from household allocation to market equilibrium over wage-plus-amenity bundles.
- The same amenity can be worth more to workers facing care constraints, long commutes, or mobility frictions.

Jobs as wage-plus-amenity bundles

$$V_i(w, a) = v_i(w) + \psi_i(a)$$

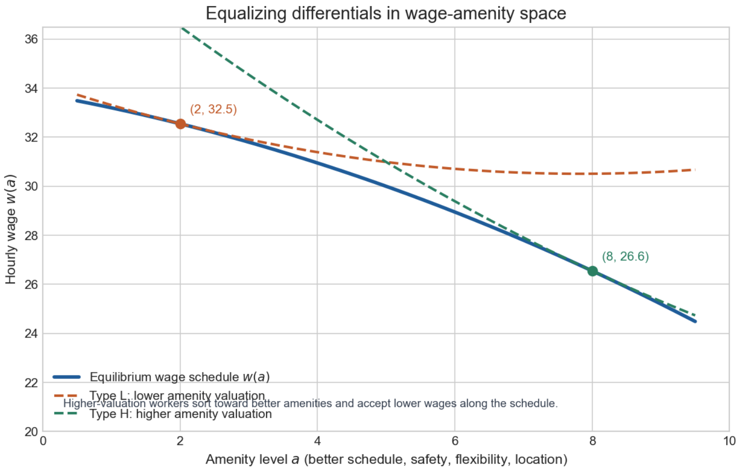
- Workers choose jobs, not wages in isolation.
- If a is a good, like schedule control or safety, workers may accept lower wages to get more of it.
- If a is a bad, like injury risk or unpredictability, wages must compensate at the margin.

Rosen hedonic equilibrium

$$\frac{dw(a)}{da} = -\frac{V_a(w,a)}{V_w(w,a)}$$

- The slope of the wage schedule equals the worker's marginal willingness to trade wages for amenities.
- Better amenities can come with lower wages in equilibrium because workers value them directly.
- Observed slopes already reflect sorting across workers with different tastes and outside options.

Hedonic wage-amenity equilibrium



Firm-side amenity provision

$$\pi_j(a) = p_j f_j(n_j, a) - w(a)n_j - C_j(a)n_j$$

- Amenities are not free: safety, staffing slack, remote infrastructure, and schedule control all have costs or productivity effects.
- Firms differ in $C_j(a)$, so the supply side of job quality is heterogeneous.
- The hedonic schedule reflects worker demand for amenities and firm costs of supplying them.

Why hedonic wage regressions can fail

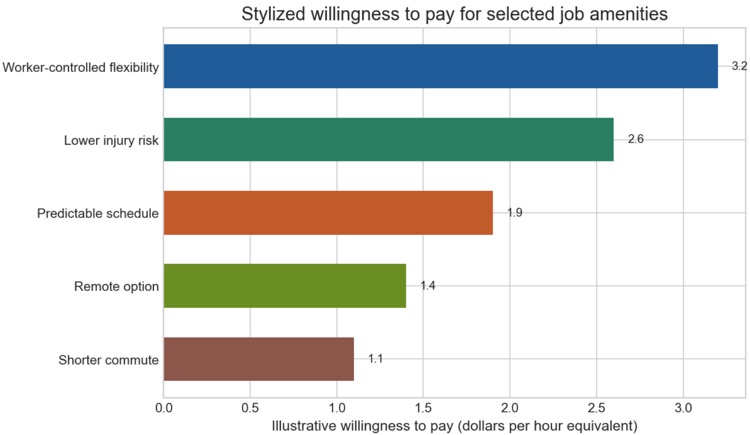
- Amenities are multidimensional and often measured badly.
- Workers sort on unobserved taste, skill, family constraints, and outside options.
- Firms bundle wages with multiple amenities and different productivity environments.
- Search frictions, mobility costs, and imperfect information break the clean frictionless benchmark.

Modern willingness-to-pay object

$$U_{ij} = \beta_w w_{ij} + A'_{ij}\beta_a + X'_{ij}\gamma + \varepsilon_{ij}$$

- Discrete-choice and conjoint designs estimate marginal valuations of named job attributes.
- Willingness to pay for amenity k is $-\beta_{a,k}/\beta_w$.
- Mas and Pallais is the Week 7 anchor because it measures explicit tradeoffs rather than inferring them from wage cross sections.

Stylized willingness to pay for selected amenities



Conceptual magnitudes anchored to modern amenity valuation evidence rather than a single paper.

Amenity taxonomy

Amenity class	Typical examples	Why it matters	Main challenge
Risks and physical conditions	Injury risk, exposure, heat, noise	Canonical compensating-differentials object	Risk mismeasurement and selection
Schedule and timing	Predictability, advance notice, nights, overtime control	Family compatibility and stress	Bundled with occupation and firm type
Location and commute	Remote work, hybrid, travel, commute time	Effective value of wages and time	Commute and remote options often missing
Autonomy and monitoring	Break control, algorithmic monitoring, discretion	Utility and productivity both move	Hard to separate from firm quality
Work intensity and pace	Pace pressure, staffing ratios, emotional load	Turnover, burnout, effort	Often poorly measured
Prestige and social environment	Mission, culture, respect, coworker quality	Sorting and welfare	Highly subjective across workers

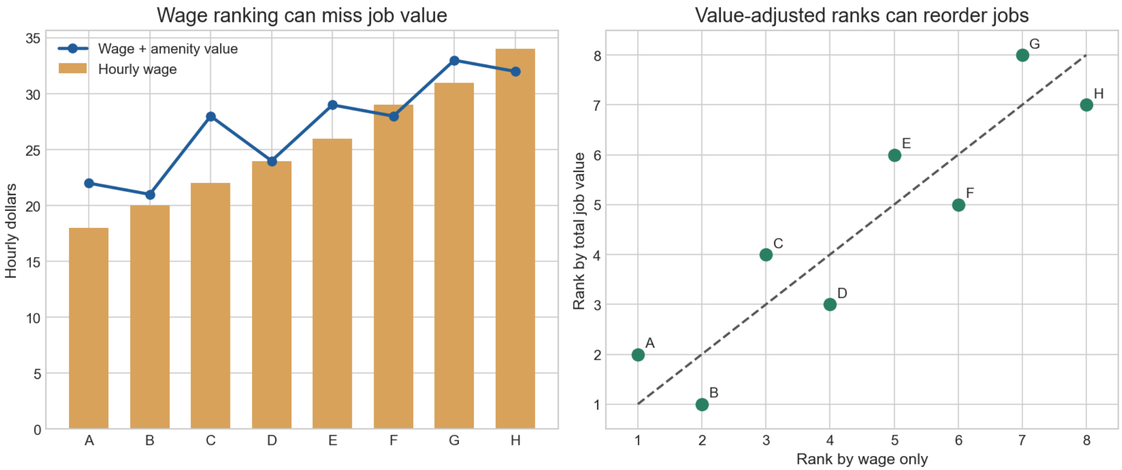
Empirical design map

Design	Main identifying object	Strength	Principal threat
Hedonic wage regression	Cross-sectional wage-amenity slope	Broad and simple	Omitted tastes, firm heterogeneity, measurement error
Risk-premium studies	Compensation for injury or fatality risk	Classic policy relevance	Risk measurement and selection
Discrete-choice / conjoint	Willingness to pay for explicit attributes	Direct valuation of named amenities	External validity and design dependence
Worker-flow / revealed preference	Latent firm or job value from mobility	Uses actual choices	Rich data and stronger structure needed
Working-conditions bundles	Value of broader job-quality bundles	Connects to inequality	Comparability across conditions

Modern amenities, sorting, and inequality

- Maestas et al. broaden the object from a few amenities to richer working-conditions bundles.
- Wage inequality can overstate or understate total-compensation inequality depending on sorting patterns.
- High-wage firms may also offer better schedules, autonomy, and safety, which amplifies inequality in job value.

Wage inequality versus total job value



Bridge to Week 8 and Week 9

- Week 8: wage dispersion is not the same as inequality in total job value once amenities are priced.
- Week 9: discrimination and segmentation may operate through access to safe, predictable, flexible, or remote jobs as well as through wages.
- Equalizing differences are therefore central to distributional interpretation, not a side topic.

Takeaways

- Rosen's equalizing-differences framework is the benchmark for wage-amenity tradeoffs.
- Naive hedonic regressions are fragile because the theory predicts sorting, heterogeneity, and bundling.
- Modern discrete-choice and worker-flow evidence recover job value more directly.
- Week 7 is the hinge from wage determination to inequality and segmentation.

Appendix: optional extension block

- Risky jobs and the value of statistical life in the spirit of Viscusi and Aldy.
- Search-friction interpretations of compensating differentials in the spirit of Hwang, Mortensen, and Reed.
- Revealed-preference firm ranking in the spirit of Sorkin.